

Transformations in space

Translation, reflection, rotation and enlargement — the plane transformations, each with one more coordinate.

LESSON 9

1 × 40 MIN

GEOMETRY 11, §5

IGCSE E7.2

LESSON CLOCK

8'

14'

12'

8'

3'

Four transformations	8 min
Coordinate rules	14 min
Isometries	12 min
Composition	8 min
Homework	3 min

LEARNING OBJECTIVES

- Apply a translation, a reflection in a coordinate plane and an enlargement in space.
- Write the coordinate rule for each transformation.
- Recognise which transformations preserve length and which do not.
- Compose two transformations.

TEXTBOOK

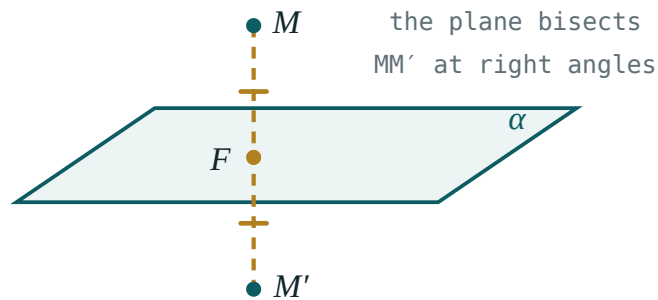
National	pp. 47–52
Cambridge	Core & Extended, pp. 349–354
Workbook	Exercise 5.1

01

Explanation

Four transformations, four rules

TRANSFORMATION	RULE	PRESERVES LENGTH?
translation by (p, q, r)	$(x + p, y + q, z + r)$	yes
reflection in $z = 0$	$(x, y, -z)$	yes
reflection in $y = 0$	$(x, -y, z)$	yes
reflection in the z -axis	$(-x, -y, z)$	yes
reflection in the origin	$(-x, -y, -z)$	yes
rotation 90° about Oz	$(-y, x, z)$	yes
enlargement, centre O , factor k	(kx, ky, kz)	no — scaled by $ k $



A point and its image in the plane $z = 0$. Only the third coordinate changes sign.

Isometries

DEFINITION

An **isometry** is a transformation preserving distance: $|A'B'| = |AB|$ for every pair. Translations, reflections and rotations are isometries; enlargements with $|k| \neq 1$ are not.

Isometries preserve everything that depends only on distance — angles, areas, volumes, congruence. An enlargement of factor k multiplies lengths by $|k|$, areas by k^2 and volumes by $|k|^3$.

A REFLECTION REVERSES ORIENTATION

A right-handed screw becomes left-handed in a mirror. That is why a reflection cannot be achieved by any rotation, however clever.

Composition

Applying two transformations in turn is itself a transformation. Two reflections in **parallel** planes give a translation of twice the gap; two reflections in **intersecting** planes give a rotation about their line of intersection, through twice the angle between them.

reflect in $z = 0$, then in $z = 3 \implies$ translation by $(0, 0, 6)$

Composition is not commutative in general: reflecting then translating is not the same as translating then reflecting.

02 Worked examples

EXAMPLE 1 Find the image of $A(2, -3, 5)$ under reflection in the plane $y = 0$, then a translation by $(1, 1, 1)$.

① Reflection: $(2, 3, 5)$.

② Translation: $(3, 4, 6)$.

ANSWER

$(3, 4, 6)$

EXAMPLE 2 A solid of volume 40 cm^3 is enlarged by factor 3. Find the new volume.

① Volumes scale by k^3 .

② 27×40

ANSWER

1080 cm^3

EXAMPLE 3 What single transformation results from reflecting in $x = 0$ then in $x = 4$?

① $(x, y, z) \rightarrow (-x, y, z)$

② $\rightarrow (8 + x, y, z)$

Reflection in $x = 4$ sends u to $8 - u$.

③ Net effect: add 8 to x .

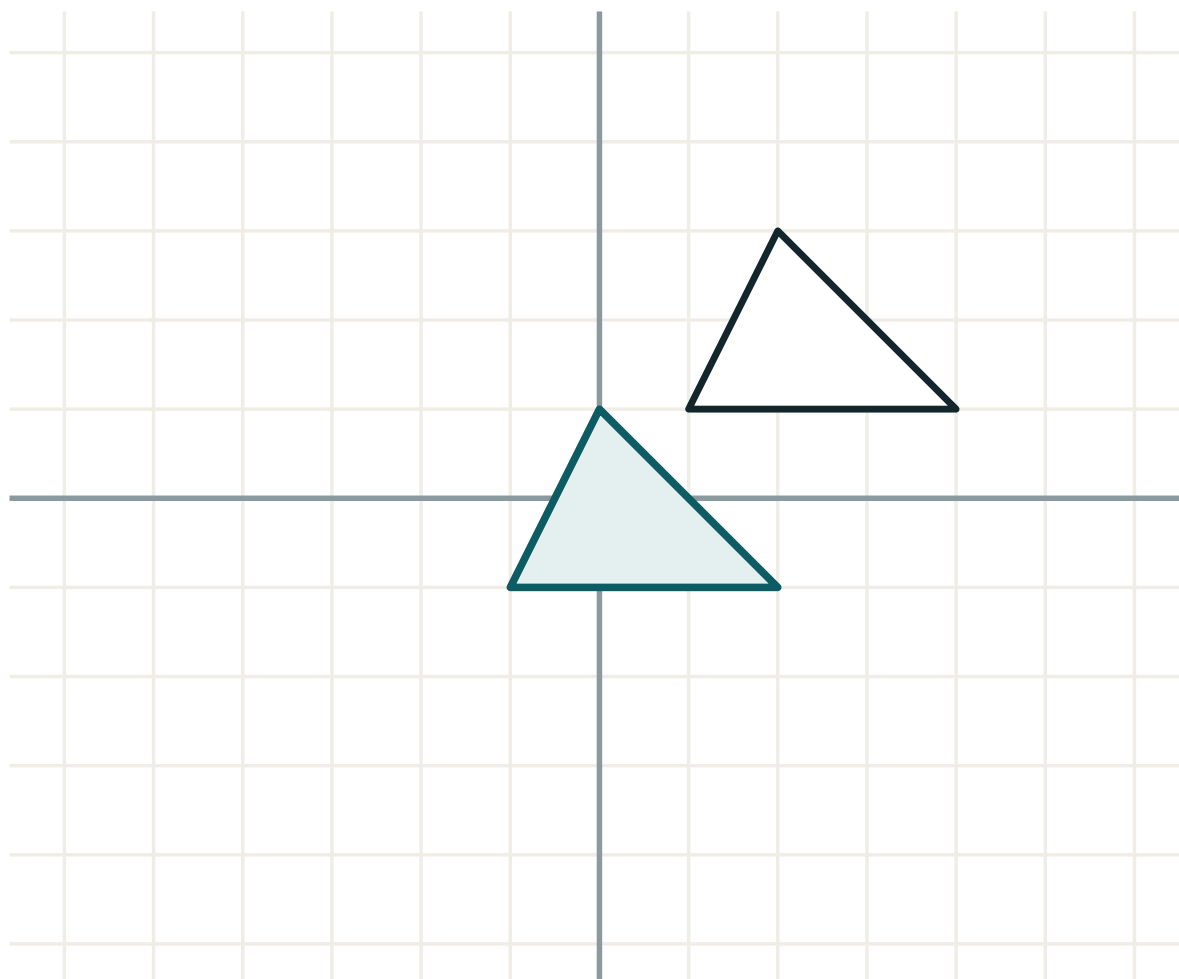
ANSWER

Translation by $(8, 0, 0)$

03 Interactive model

Reflect a shoe in a mirror to show that no rotation can reproduce a reflection.

Transforming a figure



transformation **translation**

lengths **unchanged**

angles **unchanged**

orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Transformation	Almashtirish	Преобразование
Translation	Parallel ko'chirish	Параллельный перенос
Reflection in a plane	Tekislikka nisbatan akslantirish	Отражение относительно плоскости
Rotation about an axis	O'q atrofida burish	Поворот вокруг оси
Enlargement (dilation)	Gomotetiya	Гомотетия
Centre of enlargement	Gomotetiya markazi	Центр гомотетии
Isometry	Izometriya	Изометрия
Invariant point	O'zgarmas nuqta	Неподвижная точка
Image	Tasvir	Образ

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Reflection in $z = 0$ sends (x, y, z) to:

$(-x, y, z)$

$(x, -y, z)$

$(x, y, -z)$

$(-x, -y, -z)$

An enlargement of factor 2 multiplies volume by:

2

4

8

16

Which is not an isometry?

translation

reflection

rotation

enlargement of factor 3

Two reflections in parallel planes give:

a rotation

a translation

an enlargement

the identity

Reflection in the origin sends $(1,2,3)$ to:

$(-1,2,3)$

$(1,-2,-3)$

$(-1,-2,-3)$

$(3,2,1)$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Image of $(1,2,3)$ under reflection in $z = 0$

ANSWER $(1,2,-3)$

2. Image under reflection in $x = 0$

ANSWER $(-1,2,3)$

3. Image under reflection in the origin

ANSWER $(-1,-2,-3)$

4. Image under translation $(2,2,2)$

ANSWER $(3,4,5)$

5. Image under enlargement factor 2, centre O

ANSWER $(2,4,6)$

6. Volume scale factor for $k = 3$

ANSWER 27

7. Area scale factor for $k = 4$

ANSWER 16

Medium · the standard question

7 PROBLEMS

1. Reflect $(2, -3, 5)$ in $y = 0$, then translate by $(1, 1, 1)$

ANSWER $(3, 4, 6)$

2. Rotate $(1, 0, 4)$ by 90° about Oz

ANSWER $(0, 1, 4)$

3. Two reflections in $x = 0$ and $x = 4$

ANSWER translation by $(8, 0, 0)$

4. A solid of volume 40 enlarged by 3 — new volume

ANSWER 1080

5. Surface area 24 enlarged by 5 — new area

ANSWER 600

6. Image of $(4, -2, 6)$ under enlargement factor $-\frac{1}{2}$

ANSWER $(-2, 1, -3)$

7. Is reflection an isometry?

ANSWER yes

Hard · stretch and reason

7 PROBLEMS

1. Find the reflection of $(1,2,3)$ in the plane $z = 5$

ANSWER $(1,2,7)$

2. Find the reflection of $(3,1,2)$ in the plane $x = -1$

ANSWER $(-5,1,2)$

3. Two reflections in planes at 30° — what results?

ANSWER A rotation of 60°

4. A cube of side 4 is enlarged so its volume becomes 1728. Find k

ANSWER $k = 3$

5. Show that composing two translations gives a translation

ANSWER Add the vectors

6. Why can no rotation reproduce a reflection?

ANSWER Rotations preserve orientation; reflections reverse it

7. Find the centre of an enlargement mapping $(2,2,2)$ to $(6,6,6)$ with $k = 3$

ANSWER $(0,0,0)$

07 Homework

Homework — 4 tasks

Write the coordinate rule beside every image you find.

- Find the image of $(3, -1, 4)$ under reflection in each of the three coordinate planes.
- Find the image of $(2, 5, -1)$ under enlargement of factor -2 about the origin.
- A solid has volume 54 cm^3 and surface area 90 cm^2 . Find both after an enlargement of factor 2.
- What single transformation is equivalent to reflecting in $y = 0$ and then in $y = 3$?